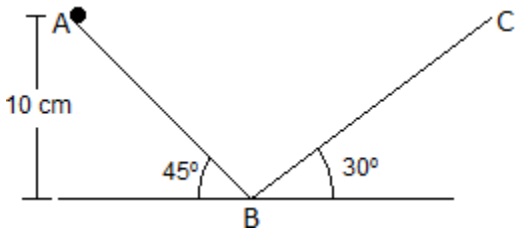
PHY101: Introduction to Physics-1

Monsoon, 2018

Tutorial-7

1. Find the time period of the motion of the particle shown in figure. Neglect the effect of the bend at the bottom. Use g = 10 ms^{-2}

(Ans: 0.97 s)



2. A 1.0 kg glider attached to a spring with a force constant of 25.0 N/m oscillates on a horizontal, frictionless air track. At t = 0 the glider is released from rest at x = -3.00 cm. (That is, the spring is compressed by 3.00 cm.) Find (a) the period of its motion, (b) the maximum values of its speed and acceleration, and (c) the position, velocity, and acceleration as functions of time.
(Ans: (a) 1.26 s, (b) 0.15 m/s, 0.75 m/s^2, (c) .....)